



**BAULKHAM HILLS  
HIGH  
SCHOOL**

**2018**

**YEAR 12  
TERM 2  
ASSESSMENTS**

# Mathematics

## Extension 1

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### **General Instructions**

- Reading time – 5 minutes
- Working time – 60 minutes
- Write using black or blue pen  
Black pen is preferred
- Board-approved calculators may be used
- Show all relevant mathematical reasoning and/or calculations
- Marks may be deducted for careless or badly arranged work
- A reference sheet is provided at the back of this paper

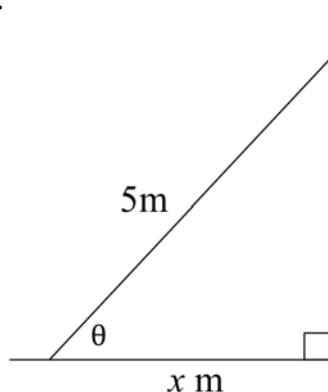
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**Total marks:  
41**

**Attempt all Questions 1 – 4 (pages 2-5)**

**Question 1 (9 Marks) Start on the appropriate page of your answer booklet.**

- a) Find the value of the constant term in the expansion of  $\left(3x^2 + \frac{2}{x}\right)^{12}$ . 2
- b) The motion of an object is defined by the equation  $x = 4\sin(\pi t)$  where  $x$  is the displacement from the origin in centimetres at time  $t$  seconds.
- (i) Write down the amplitude of the motion. 1
  - (ii) What is the period of the motion in seconds? 1
  - (iii) What is the greatest speed of the object? 1
  - (iv) Prove that the object is moving in Simple Harmonic Motion. 1
- c) A 5 metre ladder rests with one end against a wall and the other end on horizontal ground which is level with the base of the wall and  $x$  metres from the bottom of the wall. The end which is in contact with the ground is sliding away from the wall at a constant rate of 0.1 m/s.



Find the rate (in radians/s) at which the angle,  $\theta$ , between the ladder and the ground is decreasing when the end of the ladder is 3 metres from the wall. 3

**End of Question 1**

**Question 2 (13 Marks) Start on the appropriate page of your answer booklet.**

- a) The acceleration of a particle undergoing Simple Harmonic Motion is given by  $\ddot{x} = -n^2x$ , where  $x$  is the displacement of the particle from the origin in metres, and  $n$  is a positive constant. The particle starts from rest at a distance of 10 metres to the right of its centre of oscillation  $O$ . The period of the motion is 2 seconds.
- (i) Prove that  $v^2 = \pi^2(100 - x^2)$ , where  $v$  is the velocity of the particle in metres per second. **3**
- (ii) Find the speed of the particle in metres per second when it is 6 metres from the origin. **1**
- (iii) Find the time taken by the particle to first reach the point 6 metres to the right of  $O$ , correct to 2 decimal places. (You may assume  $x = a \cos nt$  where  $a$  is a constant is a solution of  $\ddot{x} = -n^2x$ ). **2**
- b) The rate of growth of a population  $N$  over  $t$  years is given by:  $\frac{dN}{dt} = -k(N - 700)$ .
- (i) Show that  $N = 700 + Ae^{-kt}$  satisfies  $\frac{dN}{dt} = -k(N - 700)$  where  $A$  and  $k$  are constants. **1**
- (ii) The population decreased from an initial population of 8300 to 5100 in 5 years. Find the population at the end of the next 5 years. Give your answer correct to the nearest hundred. **3**
- c) Consider the binomial expansion  $1 + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n}x^n = (1 + x)^n$ .
- (i) Show that  $1 + \binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{n} = 2^n$ . **1**
- (ii) Show that 
$$\left[1 + \binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{n}\right] \left[1 + \frac{1}{2}\binom{n}{1} + \frac{1}{4}\binom{n}{2} + \dots + \frac{1}{2^n}\binom{n}{n}\right] = 3^n.$$
 **2**

**End of Question 2**

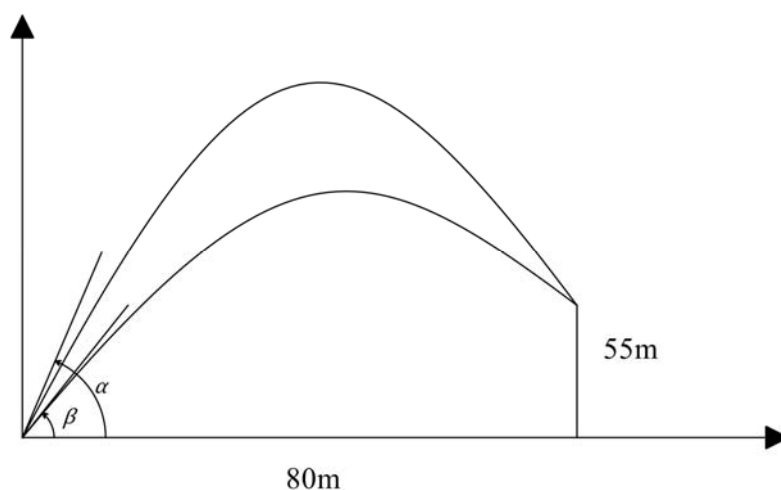
**Question 3 (9 Marks) Start on the appropriate page of your answer booklet.**

- a) The coefficients of  $x^4$  and  $x^5$  in the expansion of  $(2 + 5x)^n$  are in the ratio of 1:3      2  
Find the value of  $n$ .
- b) A certain particle moves along the  $x$  axis according to the equation  $t = 2x^2 - 5x + 3$ ,  
where  $x$  is its displacement measured in centimetres at time  $t$  seconds.  
Initially the particle is 1.5 cm to the right of the origin,  $O$ , and moving away  
from  $O$ .
- (i) Prove that the velocity,  $v \text{ cm s}^{-1}$ , is given by  $v = \frac{1}{4x - 5}$ .      2
- (ii) Find the velocity of the particle when  $t = 6$  seconds.      3
- (iii) Find an expression for the acceleration,  $a \text{ cm s}^{-2}$ , in terms of  $x$ .      2

**End of Question 3**

**Question 4 (10 Marks) Start on the appropriate page of your answer booklet.**

- a) A missile is fired from ground level into the air at a velocity of 40m/s and at an angle of  $\alpha$  with the horizontal.



A short time later another missile is fired from the same point and with the same speed but at a different angle  $\beta$ . Both missiles hit the same target at the same time. The target is 55m above the ground and 80m horizontally from the point of firing. (Take  $g = 10\text{m/s}^2$  and neglect air resistance).

Given the equations of motion at time  $t$  seconds are:

$$x = 40t\cos\alpha \text{ and } y = -5t^2 + 40t\sin\alpha \text{ (DO NOT PROVE)}$$

- (i) Show that the path of the first missile is given by  $y = x\tan\alpha - x^2\left(\frac{\sec^2\alpha}{320}\right)$ . 2
- (ii) Find the values of  $\tan\alpha$  and  $\tan\beta$ . 2
- (iii) Determine the time difference between the firing times of the two missiles. 2

- b) Consider the expansion  $(1-x)^n = \binom{n}{0} - \binom{n}{1}x + \binom{n}{2}x^2 - \dots + (-1)^n\binom{n}{n}x^n$ .

- (i) Show that  $\int_0^1 (1-x)^n dx = \binom{n}{0} - \frac{1}{2}\binom{n}{1} + \frac{1}{3}\binom{n}{2} - \dots + \frac{(-1)^n}{n+1}\binom{n}{n}$ . 2

- (ii) Hence find the value of  $\sum_{r=0}^{2018} (-1)^r \frac{1}{r+1} \binom{2018}{r}$ . 2

**End of Paper**

# HSC EXTENSION 1 TASK 3 JUNE 2018

Q1 a)

$$\begin{aligned}\text{General term: } & {}^{12}C_k (3k)^{12-k} (2k)^k \\ &= {}^{12}C_k 3^{12-k} 2^k k^{24-2k} \\ &= {}^{12}C_k 3^{12-k} 2^k k^{24-3k}\end{aligned}$$

(2) correct answer

(1) finds general term  
NB unsimplified is acceptable

For constant term  $24-3k=0$

$$k=8$$

$$\therefore \text{Constant term is } {}^{12}C_8 3^4 2^8 (=10264320)$$

b (i) Amplitude = 4

(1) correct answer

$$\begin{aligned}\text{(ii) Period} &= \frac{2\pi}{\pi} \\ &= 2\end{aligned}$$

(1) correct answer

$$\text{(iii) } x = 4\pi \cos(\pi t)$$

(1) correct answer

$\therefore$  Greatest speed =  $4\pi$  cm/s.

$$\text{(iv) } \ddot{x} = -4\pi^2 \sin(\pi t)$$

(1) correct solution

$$\ddot{x} = -\pi^2 x \text{ which is of the form } \ddot{x} = -n^2 x$$

$\therefore$  Object is moving in SHM.

$$\text{c) } \cos \theta = \frac{x}{5}$$

$$\theta = \cos^{-1} \frac{x}{5}$$

$$\frac{d\theta}{dt} = \frac{d\theta}{dx} \cdot \frac{dx}{dt}$$

$$\frac{d\theta}{dt} = \frac{-1}{\sqrt{25-x^2}} \times 0.1$$

$$\text{When } x=3 \quad \frac{d\theta}{dt} = \frac{-1}{\sqrt{16}} \times \frac{1}{10}$$

$$= -\frac{1}{40} \text{ radians/sec.}$$

(3) correct solution

(2) finds  $\frac{d\theta}{dx}$  or  $\frac{dx}{d\theta}$

(1) establishes correct chain rule

$$2) a) (i) \text{ Period} = \frac{2\pi}{n}$$

$$2 = \frac{2\pi}{n}$$

$$n = \pi$$

$$\ddot{x} = -\pi^2 x$$

$$\frac{d}{dx} \left( \frac{1}{2} v^2 \right) = -\pi^2 x$$

$$\left[ \frac{1}{2} v^2 \right]_0^x = -\pi^2 \left[ \frac{x^2}{2} \right]_0^x$$

$$\frac{1}{2} (v^2 - 0) = -\pi^2 \left( \frac{x^2}{2} - \frac{10^2}{2} \right)$$

$$\frac{v^2}{2} = \pi^2 \left( \frac{100 - x^2}{2} \right)$$

$$v^2 = \pi^2 (100 - x^2)$$

③ correct solution

② finds  $\frac{1}{2} v^2$  in terms of  $x$  (including constant)

① finds the value of  $n$   
OR ① attempts to integrate  
 $\frac{d}{dx} \left( \frac{1}{2} v^2 \right) = -n^2 x$

$$(ii) \text{ When } x=4, v^2 = \pi^2 (100 - 36)$$

$$v^2 = 64\pi^2$$

$$v = \pm \pi \sqrt{64}$$

$$\therefore \text{Speed} = 8\pi \text{ m/s}$$

① correct answer

$$(iii) \text{ When } x=6, 6 = 10 \cos \pi t$$

$$\frac{3}{5} = \cos \pi t$$

$$\pi t = 0.92729 \dots$$

$$\therefore t = 0.295167 \dots$$

$$t = 0.30 \text{ seconds (2dp)}$$

② correct answer

① substitutes  $x=6$  into

$$x = 10 \cos \pi t$$

2(b)(i)

$$N = 700 + Ae^{-kt}$$

$$\frac{dN}{dt} = -k(700 + Ae^{-kt} - 700)$$

$$= -k(N - 700)$$

① correct solution

ii) When  $t=0$ ,  $N=8300$

$$8300 = 700 + Ae^0$$

$$A = 7600$$

$$\therefore N = 700 + 7600e^{-kt}$$

When  $t=5$ ,  $N=5100$

$$5100 = 700 + 7600e^{-5k}$$

$$4400 = 7600e^{-5k}$$

$$\frac{11}{19} = e^{-5k}$$

$$-5k = \ln \frac{11}{19}$$

$$5k = \ln \frac{19}{11}$$

$$k = \frac{\ln \frac{19}{11}}{5}$$

$$(\approx 0.1093...)$$

③ correct answer

② finds  $A$  and  $k$

① finds correct value of  $A$  and attempts to find  $k$

$$\text{When } t=10, N = 700 + 7600e^{-10 \times \frac{\ln \frac{19}{11}}{5}}$$

$$= 3247.368...$$

$\therefore$  Population is approximately 3200

2(c) i) Let  $x=1$

$$1 + \binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{n} = (1+1)^n$$

$$1 + \binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{n} = 2^n$$

① correct solution

$$\text{ii) let } x = \frac{1}{2} \text{ in } 1 + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n}x^n = (1+x)^n \quad \text{② correct solution}$$

$$\therefore 1 + \frac{1}{2}\binom{n}{1} + \frac{1}{4}\binom{n}{2} + \dots + \frac{1}{2^n}\binom{n}{n} = (1 + \frac{1}{2})^n \quad \text{① substitutes } x = \frac{1}{2}$$

$$\left[1 + \binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{n}\right] \left[1 + \frac{1}{2}\binom{n}{1} + \frac{1}{4}\binom{n}{2} + \dots + \frac{1}{2^n}\binom{n}{n}\right] = 2^n \times 1.5^n = (2 \times 1.5)^n$$

$$= 3^n$$

3 (a) General term:  ${}^nC_k 2^{n-k} (5x)^k$

$$\therefore \frac{{}^nC_4 2^{n-4} 5^4}{{}^nC_5 2^{n-5} 5^5} = \frac{1}{3}$$

$$\frac{2}{5} \frac{\frac{n!}{(n-4)!4!}}{n!} = \frac{1}{3}$$

$$\frac{(n-5)!}{(n-4)(n-5)!} \times 8 = \frac{5}{6}$$

$$n-4=6$$

$$n=10$$

② correct answer

① equates ratio of coefficients to  $\frac{1}{3}$ .

b) i)  $t = 2x^2 - 5x + 3$

$$\frac{dt}{dx} = 4x - 5$$

$$\frac{dx}{dt} = \frac{1}{4x-5}$$

② correct solution

① finds  $\frac{dt}{dx}$

(ii) When  $t=6$ ,  $6 = 2x^2 - 5x + 3$

$$2x^2 - 5x - 3 = 0$$

$$(2x+1)(x-3) = 0$$

$$x = -\frac{1}{2}, x = 3$$

But when  $t=0$ ,  $x = 1.5$  and  $v = \frac{1}{1} = 1 \text{ cm/s}$

Since  $v \neq 0$ , particle never stops and can't reach  $\frac{1}{2}$ .

$$\therefore v = \frac{1}{4x-5} = \frac{1}{-7} \text{ cm/s}$$

③ correct solution

② finds  $v = \pm \frac{1}{7}$

① calculates  $x=3$

iii)  $a = \frac{d}{dx} \left( \frac{1}{2} v^2 \right)$

$$= \frac{1}{2} \frac{d(4x-5)^{-2}}{dx}$$

$$= \frac{1}{2} \times -2 \times 4 (4x-5)^{-3}$$

$$a = \frac{-4}{(4x-5)^3}$$

② correct solution

① attempts to differentiate

$$\frac{d}{dx} \left( \frac{1}{2} v^2 \right)$$

4 a)(i) From  $x = 40t \cos \alpha$

$$t = \frac{x}{40 \cos \alpha}$$

sub in  $y = -5t^2 + 40t \sin \alpha$

$$y = -5 \frac{x^2}{1600 \cos^2 \alpha} + \frac{40}{40 \cos \alpha} x \sin \alpha$$

$$y = x \tan \alpha - \frac{x^2}{320} \sec^2 \alpha$$

(ii) When  $x = 80$   $y = 55$

$$55 = 80 \tan \alpha - \frac{6400 \sec^2 \alpha}{320}$$

$$55 = 80 \tan \alpha - 20(\tan^2 \alpha + 1)$$

$$20 \tan^2 \alpha - 80 \tan \alpha + 175 = 0$$

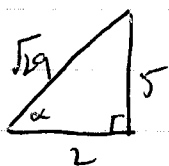
$$4 \tan^2 \alpha - 16 \tan \alpha + 115 = 0$$

$$(2 \tan \alpha - 3)(2 \tan \alpha - 5) = 0$$

$$\tan \alpha = \frac{3}{2}, \tan \alpha = \frac{5}{2}$$

$$\therefore \tan \alpha = \frac{5}{2} \text{ \& \; } \tan \beta = \frac{3}{2}$$

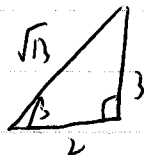
iii)



$$x = 40t \cos \alpha$$

$$80 = 40t \frac{2}{\sqrt{29}}$$

$$t = \sqrt{29}$$



$$x = 40t \cos \beta$$

$$80 = 40t \frac{2}{\sqrt{13}}$$

$$t = \sqrt{13}$$

$$\therefore \text{Difference in time} = \sqrt{29} - \sqrt{13}$$

$$= 1.7796 \dots \text{ sec}$$

(2) correct solution

(1) correctly attempts to eliminate  $t$  from displacement equations

(2) correct solution

(must specify  $\tan \alpha = \frac{5}{2}$  and  $\tan \beta = \frac{3}{2}$  as  $\alpha > \beta$ )

(1) quadratic in  $\tan \alpha$

(2) correct answer

(1) calculates time for one of the missiles

4 iii) When  $x = 80, y = 55$

$$55 = 80 \tan \alpha - 6400 \frac{\sec^2 \alpha}{320}$$

$$55 = 80 \tan \alpha - 20(\tan^2 \alpha + 1)$$

$$20 \tan^2 \alpha - 80 \tan \alpha + 75 = 0$$

$$4 \tan^2 \alpha - 16 \tan \alpha + 15 = 0$$

$$(2 \tan \alpha - 3)(2 \tan \alpha - 5) = 0$$

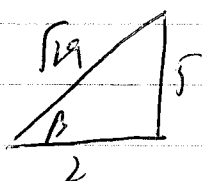
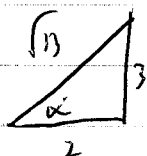
$$\tan \alpha = \frac{3}{2} \quad \tan \alpha = \frac{5}{2}$$

$$\therefore \tan \alpha = \frac{3}{2} \text{ \& } \tan \beta = \frac{5}{2}$$

② correct solution

① solves quadratic in  $\tan \alpha$

iv)



$$x = 40 \cos \alpha$$

$$80 = 40 \cos \frac{2}{\sqrt{13}}$$

$$t = \sqrt{13}$$

$$\therefore \text{difference} = \sqrt{29} - \sqrt{13}$$

$$= 1.7796 \dots \text{secs}$$

$$x = 40 \cos \beta$$

$$80 = 40 \cos \frac{2}{\sqrt{29}}$$

$$t = \sqrt{29}$$

② correct answer

① calculates  $x$

① calculates time for 1 missile

$$\begin{aligned}
 4b) i) \int_0^1 (1-x)^n dx &= \int_0^1 \left( \binom{n}{0} - \binom{n}{1}x + \binom{n}{2}x^2 + \dots + (-1)^n \binom{n}{n}x^n \right) dx \\
 &= \left[ \binom{n}{0}x - \binom{n}{1}\frac{x^2}{2} + \binom{n}{2}\frac{x^3}{3} + \dots + (-1)^n \binom{n}{n}\frac{x^{n+1}}{n+1} \right]_0^1 \\
 &= \left( \binom{n}{0} - \binom{n}{1}\frac{1}{2} + \binom{n}{2}\frac{1}{3} + \dots + (-1)^n \binom{n}{n}\frac{1}{n+1} \right) - (0 - 0 + 0 + \dots + 0) \\
 &= \binom{n}{0} - \frac{1}{2}\binom{n}{1} + \frac{1}{3}\binom{n}{2} + \dots + \frac{(-1)^n}{n+1}\binom{n}{n} \quad \textcircled{2} \text{ correct solution} \\
 &\quad \textcircled{1} \text{ Integrates RHS}
 \end{aligned}$$

$$\begin{aligned}
 b(ii) \int_0^1 (1-x)^{2018} dx &= \left[ -\frac{(1-x)^{2019}}{2019} \right]_0^1 \\
 &= 0 + \frac{1}{2019}
 \end{aligned}$$

\textcircled{2} correct solution  
 \textcircled{1} integrates  $\int_0^1 (1-x)^n dx$

$$\begin{aligned}
 \therefore \text{from (i)} \quad \binom{2018}{0} - \frac{1}{2}\binom{2018}{1} + \frac{1}{3}\binom{2018}{2} + \dots + \frac{(-1)^{2018}}{2019}\binom{2018}{2018} &= \sum_{r=0}^{2018} (-1)^r \frac{1}{r+1} \binom{2018}{r} \\
 \therefore \sum_{r=0}^{2018} (-1)^r \frac{1}{r+1} \binom{2018}{r} &= \frac{1}{2019}
 \end{aligned}$$